

Software Design Document (SDD)

Event Management System

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Project Title: Event Management System

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1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture and system design of the Event Management System. The document is intended for software developers, system analysts, project supervisors, testers, and end users involved in the development and implementation of the system.

The document explains how the system components, interfaces, database structures, and user interactions are designed to meet the system requirements.

1.2 Scope

The Event Management System is a web-based application designed to help organizations and individuals plan, organize, and manage events efficiently.

The system allows:

- Event creation and scheduling
- Online event registration
- Ticket booking and payment tracking
- User account management
- Notifications and reminders
- Report generation

Goals

- Simplify event planning processes
- Improve communication between organizers and attendees
- Reduce paperwork and manual processes
- Enhance event tracking and management

Benefits

- Faster event registration
- Secure storage of event data
- Improved organization and coordination
- Better attendee experience

1.3 Overview

This document is organized into several sections describing the overall system architecture, data structures, component designs, interface designs, and requirement mappings of the Event Management System.

1.4 Reference Material

The following materials were used:

- IEEE Std 1016-2009 Software Design Description
- Software Engineering Textbooks
- Database Design Principles
- Web Application Development Guidelines



1.5 Definitions and Acronyms

Term/Acronym	Meaning
EMS	Event Management System
UI	User Interface
DBMS	Database Management System
Admin	System Administrator
User	Registered system participant
API	Application Programming Interface

2. SYSTEM OVERVIEW

The Event Management System is designed to automate event planning and management activities.

The system supports three types of users:

- . Administrator
- . Event Organizer
- . Attendee

Main Functions

- User registration and login
- Event creation and editing
- Ticket booking
- Event search and filtering
- Payment processing
- Notification management
- Reporting and analytics

The system will operate through a web browser and use a centralized database for storing all information.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

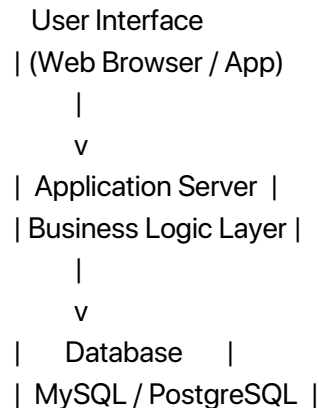
The system follows a three-tier architecture:

- . Presentation Layer
- . Application Layer
- . Database Layer



Architecture Diagram

Plaintext



Subsystems

- Authentication Subsystem
- Event Management Subsystem
- Booking Subsystem
- Payment Subsystem
- Notification Subsystem
- Reporting Subsystem

3.2 Decomposition Description

Functional Modules

1. User Management Module

- *Functions:* Register user, Login user, Manage profiles, Reset password

2. Event Management Module

- *Functions:* Create event, Update event, Delete event, View event details

3. Booking Module

- *Functions:* Book tickets, Cancel booking, Confirm attendance

4. Payment Module

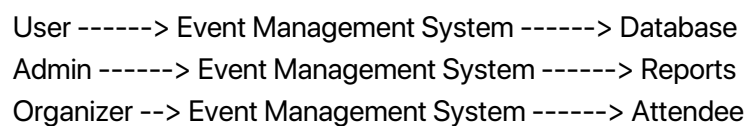
- *Functions:* Process payments, Generate receipts, Verify transactions

5. Notification Module

- *Functions:* Send email notifications, Send reminders, Notify organizers

Data Flow Diagram (DFD)

Plaintext



3.3 Design Rationale

The three-tier architecture was selected because:

- It improves scalability
- It supports easier maintenance



- It separates business logic from the user interface
- It improves security and performance

Alternative architectures such as monolithic systems were considered but rejected due to poor scalability and maintenance difficulties.

4. DATA DESIGN

4.1 Data Description

The system uses a relational database to store and manage information.

Main Entities

- Users
- Events
- Bookings
- Payments
- Notifications

Database Tables

- . Users Table
- . Events Table
- . Bookings Table
- . Payments Table
- . Notifications Table

4.2 Data Dictionary

Entity	Attribute	Type	Description
User	UserID	Integer	Unique user identifier
User	Name	Varchar	Full name
User	Email	Varchar	User email
Event	EventID	Integer	Unique event identifier
Event	Title	Varchar	Event title
Event	Date	Date	Event date
Booking	BookingID	Integer	Booking identifier
Booking	TicketCount	Integer	Number of tickets
Payment	PaymentID	Integer	Payment identifier
Payment	Amount	Decimal	Payment amount

5. COMPONENT DESIGN

1. User Login Function

Plaintext

BEGIN

INPUT username

INPUT password

IF username and password are valid THEN



```

    ALLOW access
ELSE
    DISPLAY error message
ENDIF
END
2. Event Creation Function
Plaintext
BEGIN
    INPUT event details

    VALIDATE event information

    IF information is valid THEN
        SAVE event to database
        DISPLAY success message
    ELSE
        DISPLAY validation error
    ENDIF
END

```

3. Ticket Booking Function

```

Plaintext
BEGIN
    SELECT event
    INPUT number of tickets

    CHECK ticket availability

    IF tickets available THEN
        PROCESS booking
        GENERATE receipt
    ELSE
        DISPLAY unavailable message
    ENDIF
END

```

6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The system interface will be user-friendly and responsive.

Features

- Easy navigation menus
- Search functionality
- Dashboard overview
- Event calendar



- Online booking forms

Users will receive confirmation messages after completing actions such as registration or booking.

6.2 Screen Images

Login Screen

Plaintext

```
-----
|  EVENT MANAGEMENT  |
-----
| Username: [_____] |
| Password: [_____] |
|                   |
| [ Login ] [ Register ] |
-----
```

Dashboard Screen

Plaintext

```
-----
| Dashboard          |
-----
| 1. Create Event    |
| 2. View Events     |
| 3. Manage Bookings |
| 4. Reports         |
-----
```

Booking Screen

Plaintext

```
-----
| Book Event Ticket  |
-----
| Event Name: Tech Conference |
| Tickets: [_____] |
|                   |
| [ Confirm Booking ] |
-----
```

6.3 Screen Objects and Actions

Screen Object	Action
Login Button	Authenticates user
Register Button	Creates account
Create Event Button	Opens event form
Book Ticket Button	Processes booking
Search Bar	Searches events



7. REQUIREMENTS MATRIX

Requirement ID	Requirement Description	Component
FR1	User registration	User Management Module
FR2	User login	Authentication Module
FR3	Create event	Event Management Module
FR4	Book tickets	Booking Module
FR5	Process payments	Payment Module
FR6	Generate reports	Reporting Module

8. APPENDICES

Appendix A: Tools and Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: PHP / Python / Node.js
- Database: MySQL
- Server: Apache / XAMPP

Appendix B: Future Improvements

- Mobile application support
- QR code ticket validation
- AI-based event recommendations
- Integration with online payment gateways

Sources

1. <https://slideplayer.com/slide/7009024/>
2. <https://slideplayer.com/slide/5086035/>
3. <https://slideplayer.com/slide/5086035/>

